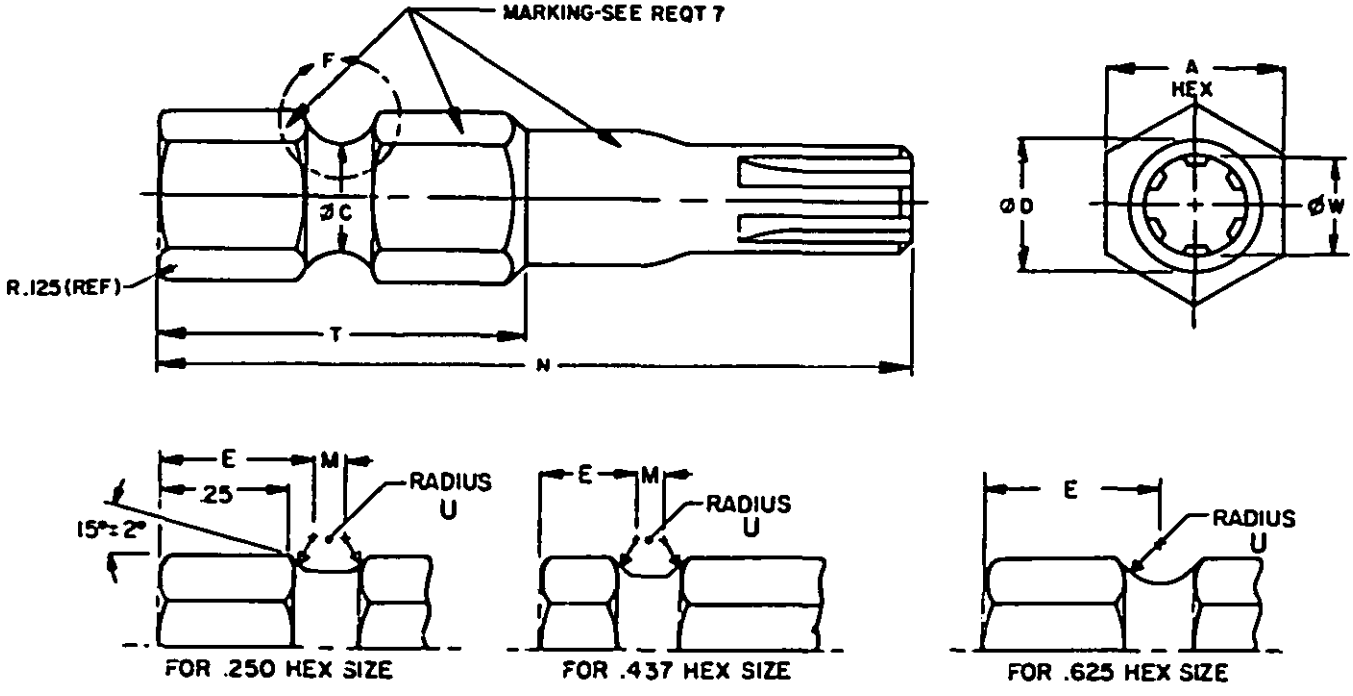
MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, SIX SPLINE, STRAIGHT, LONG EXTERNAL HEX SHANK, POWER DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the bits described herein shall consist of this specification and the latest issue of MIL-B-9946.



VIEW- F -RETAINING GROOVE

NOTES:
1. All dimensions are in inches
2. Bit, retaining groove and tip point shall be in accordance with TABLE I.

FIGURE 1. BIT, SCREWDRIVER, SIX SPLINE, STRAIGHT, LONG EXTERNAL HEX SHANK, POWER DRIVEN

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	DRIVER SIZE	ACCESS SIZE	SCREW DIAMETER SIZE RANGE FOR ACCESS SIZE REF				A DIA +.000 -.004	Ø C ±.010	Ø D REAM	E ±.020	D DIA ±.005	K ±.020	T	TIP DIMENSIONS						SECOND DASH NUMBER E ±.120 BIT LENGTH				TORSIONAL HEIGHT REF - IN
			CAP SCREW	FLAT CSK HEAD	BUTTON HEAD	SET SCREW								Ø P MAJOR MAX	Ø P MAJOR MIN	Ø W MINOR MAX	Ø W MINOR MIN	S WIDTH OF SPACE MAX	S WIDTH OF SPACE MIN	L LENGTH PENETRATE SURFACE MIN	K CHAMFER MAX	1.94	2.75	3.50	6.00	
096	.096	.096	#2 & #3	#5 & #6	#5 & #6	#8	.250	.188	.247 .242	.375	.094	.062	1.375 .625	.0960	.0950	.0790	.0775	.0240	.0230	.125	.008	A	B	C	D	29
111	.111	.111	#4 & #5	#8	#8	#10	.250	.188	.247 .242	.375	.094	.062	1.375 .625	.1110	.1100	.0940	.0925	.0280	.0270	.125	.009	A	B	C	D	50
133	.133	.133	#6	—	—	—	.250	.188	.247 .242	.375	.094	.062	1.375 .625	.1330	.1310	.1140	.1120	.0340	.0320	.156	.014	A	B	C	D	65
145	.145	.145	—	#10	#10	1/4	.250	.188	.247 .242	.375	.094	.062	1.375 .625	.1450	.1435	.1240	.1225	.0355	.0340	.187	.015	A	B	C	D	111
168	.168	.168	#8	—	—	—	.250	.188	.247 .242	.375	.094	.062	1.375 .625	.1680	.1660	.1440	.1420	.0410	.0390	.216	.016	A	B	C	D	131
183	.183	.183	#10	1/4	1/4	5/16	.250	.188	.247 .242	.375	.094	.062	1.375 .625	.1830	.1815	.1580	.1565	.0440	.0425	.216	.016	A	B	C	D	216
216	.216	.216	1/4	5/16	5/16	3/8	.250	.188	.247 .242	.375	.094	.062	1.375 .625	.2160	.2145	.1840	.1825	.0550	.0535	.250	.022	A	B	C	D	328
251	.251	.251	—	3/8	3/8	7/16	.437	.344	.434 .429	.250	.109	.047	1.375 1.000	.2510	.2495	.2140	.2125	.0655	.0640	.281	.024	A	B	C	D	530
291	.291	.291	5/16	7/16	7/16	1/2	.625	.531	.422 .417	.344	.172	—	1.375 1.000	.2910	.2895	.2460	.2445	.0775	.0760	.312	.030	A	B	C	D	1,106

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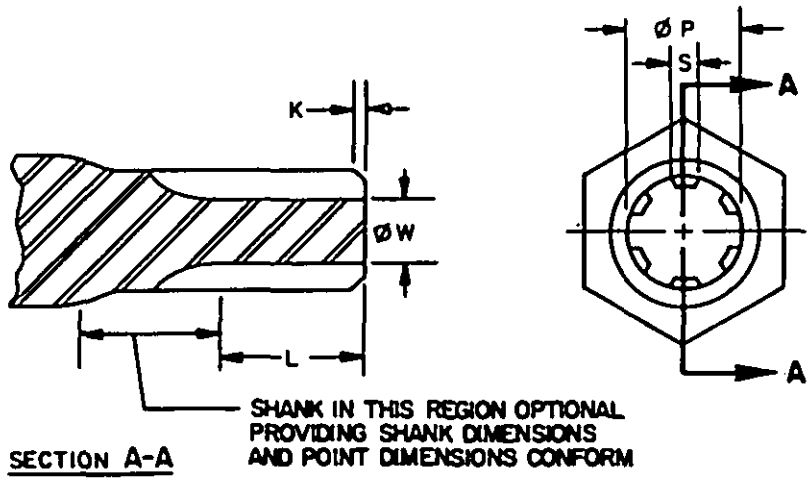


FIGURE 2. TIP POINT

REQUIREMENTS:

1. Material: Alloy Tool Steel in accordance with American Society for Testing Material (ASTM) A681, Shock-Resisting Steel, Type S-2, (UNS T41902).

2. Hardness: Macrohardness, (Rockwell C Scale) 56-62HRC. Three macrohardness readings shall be taken at the locations shown in FIGURE 3. Each hardness value must be within the range specified.

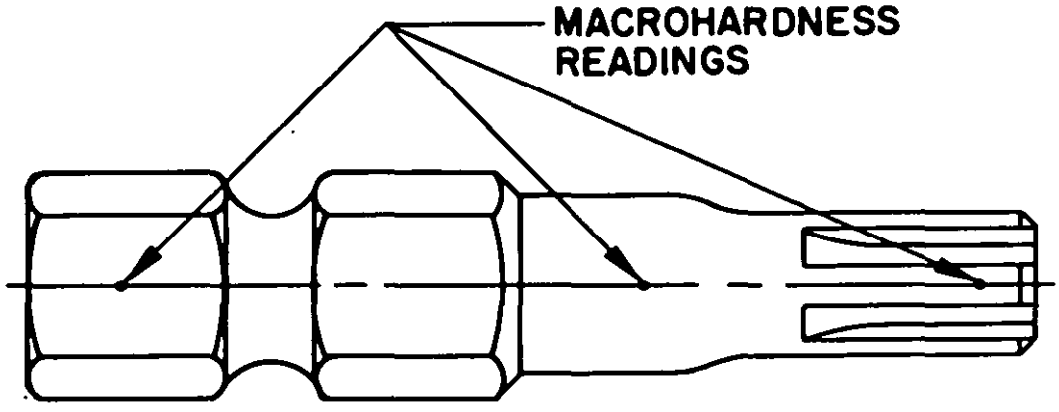


FIGURE 3. MACROHARDNESS READING LOCATIONS

Superficial hardness (Rockwell HR15N scale) 87-90. Superficial hardness readings shall be taken on bid samples at two locations on sectionalized specimens along the blade.

3. Tip point: Tip point shall be in accordance with FIGURE 2 and TABLE I.

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4. Driver bit gage: Driver bit gage shall be in accordance with FIGURE 4 and TABLE II.

5. Test block: Test block shall be in accordance with FIGURE 5 and TABLE III and bits shall not fail when tested in accordance with Requirement 5. Torque tests are conducted using this block as described herein and in MIL-B-9946.

6. Torsion test: When tested, all bits shall withstand the test load specified in TABLE I without injury or permanent deformation which may affect their durability or serviceability. A torsion test shall be conducted to determine conformance with the test load specified in TABLE I. The bit shall be inserted into a female socket designed for the size bit being tested. The bit shall withstand the torsional moment (test load) specified in TABLE I. An angular reference line shall be established and the bits shall show less than a 5° permament angular deflection. The bit shall show no flaws, cracks or other defects which might affect its durability or serviceability. The part of the bit inserted into the socket shall show no distortion or rounded corners. For bit sizes .096 inches thru .251 inches, inclusive, the torsional load shall be further increased until the bit fails. Failure shall occur by a uniform shear fracture for these sizes. Sample bits shall be tested using a test recess conforming to FIGURE 5 and TABLE III. There shall be no debris present in the recess during the test.

7. Identification marking: Driver bits shall be marked by etching, stamping or engraving onto each bit the following minimum information:

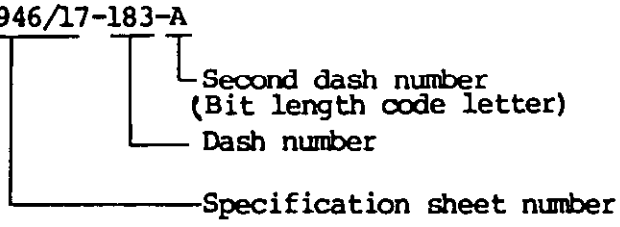
   a. Military specification part number.

   b. Manufacturer's symbol or code.

   c. Country of manufacture.

8. Part number: The part number consists of the letter "M", the basic number of this specification sheet, followed by a dash number and a code letter for bit length taken from TABLE I.

Example: M9946/17-183-A



M9946/17-183-A indicates - Bit, Screwdriver, Six Spline, Straight, Long External Hex Shank, Power Driven; Driver Size Max .183; Recess Size .183; Screw Diameter Size Ref, 10, 1/4, 1/4, and 5/16; Hex Size .250; Bit Length 1.94;

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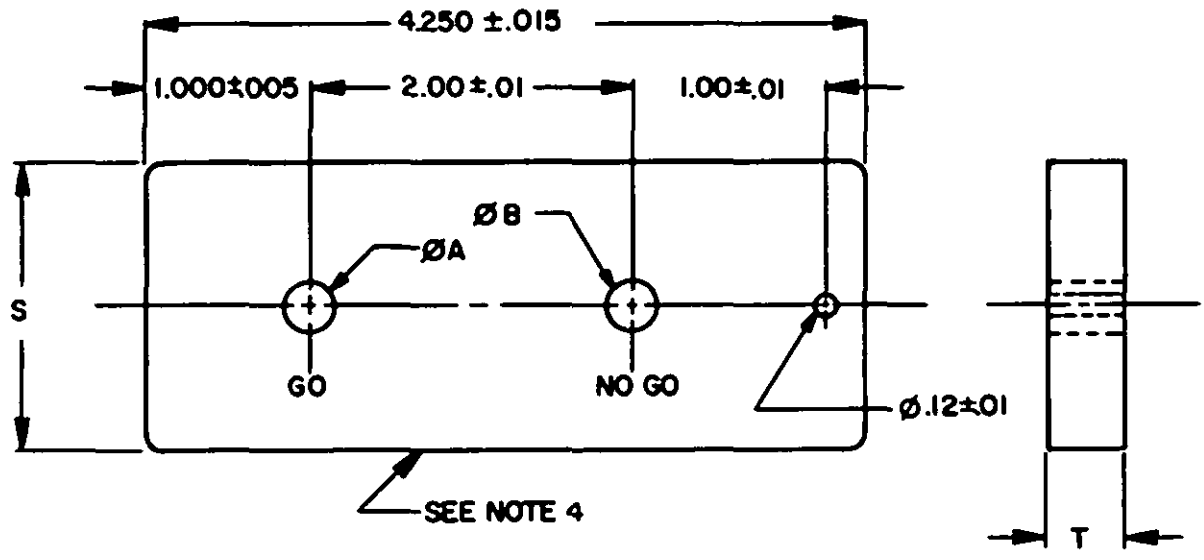


FIGURE 4. DRIVER BIT GAGE

TABLE II. DRIVER BIT GAGE DIMENSIONS FOR SIX SPLINE BITS

KEY AND SOCKET SIZE NOMINAL	GO ØA MAX	GO ØA MIN	NO GO ØB MAX	NO GO ØB MIN	S MIN	T ±.005
.096	.0963	.0960	.0950	.0947	1	.187
.111	.1113	.1110	.1100	.1097	1	.187
.133	.1333	.1330	.1310	.1307	1	.187
.145	.1453	.1450	.1435	.1432	1	.187
.168	.1683	.1680	.1660	.1657	1	.375
.183	.1833	.1830	.1815	.1812	1	.375
.216	.2163	.2160	.2145	.2142	1	.375
.251	.2513	.2510	.2495	.2492	1	.375
.291	.2913	.2910	.2895	.2892	1	.625

NOTES:
1. All dimensions are in inches.
2. Material: CRES in accordance with ASTM A276, Type 440C (UNS S44004).
3. Hardness: 58-62HRC
4. Marking: Stamp nominal key and socket size and "GO" and "NO GO".

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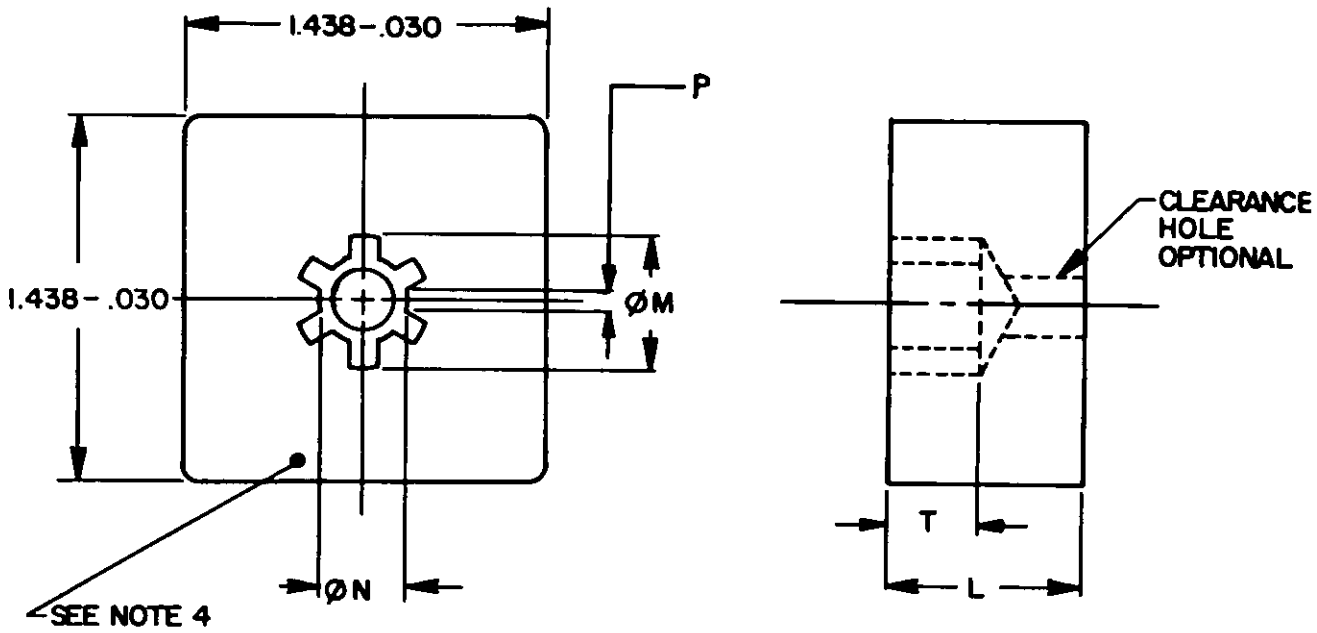


FIGURE 5. TEST BLOCK

TABLE III. TEST BLOCK DIMENSIONS FOR SIX SPLINE BITS

KEY AND SOCKET SIZE NOMINAL	ØM MAJOR		ØN MINOR		P WIDTH OF SPACE		T MINIMUM ENGAGEMENT STRAIGHT SIDES	L THICKNESS REF
	MAX	MIN	MAX	MIN	MAX	MIN		
.096	.098	.097	.082	.080	.022	.021	.096	.187
.111	.115	.113	.098	.096	.025	.023	.111	.187
.133	.137	.135	.118	.116	.030	.028	.133	.187
.145	.149	.147	.128	.126	.032	.030	.145	.187
.168	.173	.171	.150	.147	.036	.033	.168	.375
.183	.188	.186	.163	.161	.039	.037	.183	.375
.216	.221	.219	.190	.188	.050	.048	.216	.375
.251	.256	.254	.221	.219	.060	.058	.251	.375
.291	.298	.296	.254	.252	.068	.066	.291	.750

NOTES:
1. All dimensions are in inches.
2. Material: Alloy Tool Steel in accordance with ASTM A681, Type S-2 (UNS T41902).
3. Hardness: 62-64HRC
4. Marking: Stamp nominal key and socket size.

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